Q1 JEE Main 2020 - 2 September (Evening)

An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true?

(A) the mean free path of the molecules decreases.
(B) the mean collision time between the molecules decreases.
the mean free path remains unchanged.
(D) the mean collision time remains unchanged

(A) (C) and (D)
(B) (A) and (D)
(C) (B) and (C)
(D) (A) and (B)

Q2 JEE Main 2020 - 4 September (Morning)

A closed vessel contains 0.1 mole of a monatomic ideal gas at $200\mathrm{K}$. If 0.05 mole of the same gas at $400\mathrm{K}$ is added to it, the final equilibrium temperature (in K) of the gas in the vessel will be close to

Q3 JEE Main 2020 - 4 September (Evening)

The change in the magnitude of the volume of an ideal gas when a small additional pressure ΔP is applied at a constant temperature, is the same as the change when the temperature is reduced by a small quantity $\Delta \mathrm{T}$ at constant pressure. The initial temperature and pressure of the gas were $300\mathrm{K}$ and 2 atm. respectively. If $|\Delta T|=\mathrm{C}|\Delta \mathrm{P}|$ then value of C in (K/atm.) is

Q4 JEE Main 2020 - 5 September (Morning)

Number of molecules in a volume of $4\mathrm{cm}^3$ of a perfect monoatomic gas at some temperature T and at a pressure of $2\mathrm{cm}$ of mercury is close to ? (Given, mean kinetic energy of a molecule (at T) is 4×10^{-14} erg, $g=980\mathrm{cm}/\mathrm{s}^2$, density of mercury $=13.6\mathrm{g}/\mathrm{cm}^3$)

(A) 5.8×10^{18}
(B) 4.0×10^{16}
(C) 5.8×10^{16}
(D) 4.0×10^{18}

Q5 JEE Main 2020 - 5 September (Evening)

Nitrogen gas is at $300^{\circ}\mathrm{C}$ temperature. The temperature (in K) at which the rms speed of a H_2 molecule would be equal to the rms speed of a nitrogen molecule, is
(Molar mass of N_2 gas $28\mathrm{g}$).

Q6 JEE Main 2020 - 6 September (Morning)

Initially a gas of diatomic molecules is contained in a cylinder of volume V_1 at a pressure P_1 and temperature 250 K. Assuming that 25% of the molecules get dissociated causing a change in number of moles. The pressure of the resulting gas at temperature $2000\mathrm{K}$, when contained in a volume $2\mathrm{V}_1$ is given by P_2. The ratio P_2/P_1 is

Q7 JEE Main 2020 - 6 September (Evening)

In a dilute gas at pressure P and temperature T, the mean time between successive collisions of a molecule varies with T as

(A) $\sqrt{T}$
(B) T
(C) $\frac{1}{\sqrt{\mathrm{T}}}$
(D) $\frac{1}{\mathrm{T}}$

Q8 JEE Main 2020 - 6 September (Evening)

Assuming the nitrogen molecule is moving with r.m.s. velocity at $400\mathrm{K}$, the de-Broglie wavelength of nitrogen molecule is close to
(Given : nitrogen molecule weight: 4.64×10^{-26} kg, Boltzman constant : $1.38 \times 10^{-23}\mathrm{J/K}$, Planck constant : $6.63 \times 10^{-34}\mathrm{J.s}$)

(A) 0.24 Å
(B) 0.20 Å
(C) 0.34 Å
(D) 0.44 Å

Q9 JEE Main 2020 - 7 January (Morning)

Three moles of ideal gas A with $\frac{C_P}{C_V}=\frac{4}{3}$ is mixed with two moles of another ideal gas B with $\frac{C_P}{C_V}=\frac{5}{3}$. The $\frac{C_P}{C_V}$ of mixture is (Assuming temperature is constant)

(A) 1.5
(B) 1.42
(C) 1.7
(D) 1.3

Q10 JEE Main 2020 - 7 January (Morning)

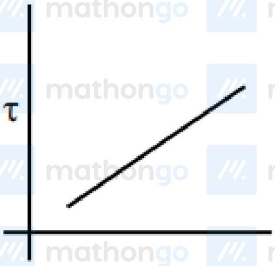
A litre of dry air at STP expands adiabatically to a volume of$3\ litres$. If $\gamma=1.40$, the work done by air is : ($3^{1.4}=4.6555$) [Take air to be an ideal gas]

(A) $100.8\ J$
(B) $90.5\ J$
(C) $45\ J$
(D) $18\ J$

Q11 JEE Main 2020 - 8 January (Morning)

The plot that depicts the behavior of the mean free time τ (time between two successive collisions) for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is: (Graphs are schematic and not drawn to scale)

(A)



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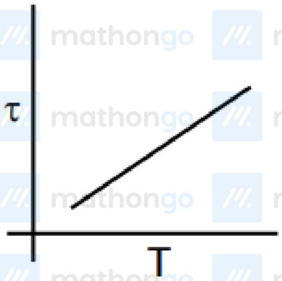

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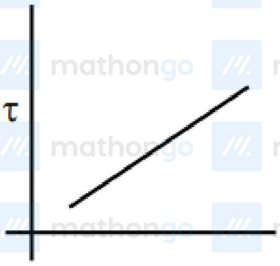

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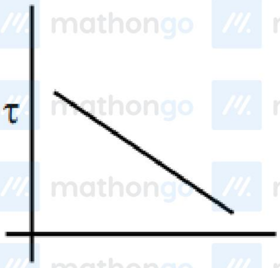
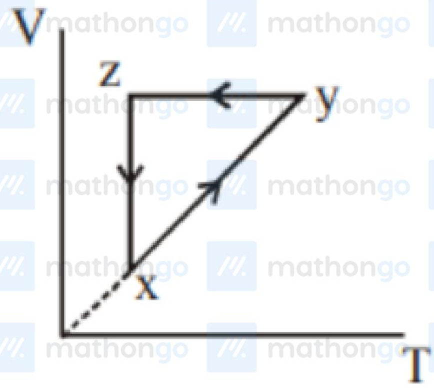
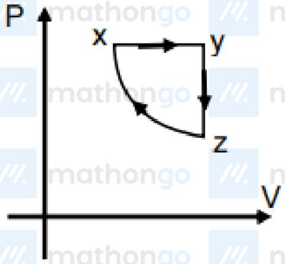


Q12 JEE Main 2020 - 8 January (Morning)

Choose the correct P-V graph of ideal gas for given V-T graph.



(A)



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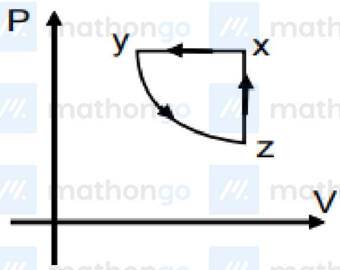

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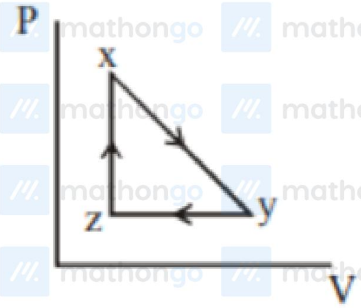
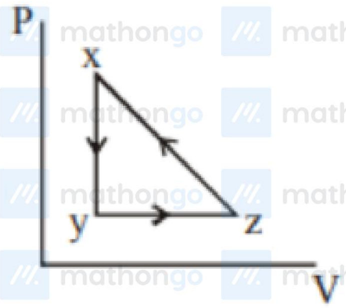


(D)



Q13 JEE Main 2020 - 8 January (Evening)

n mole of He and $2n$ mole of O_2 is mixed in a container. Then $\left(\frac{C_P}{C_V}\right)_{mix}$ will be

(A) $\frac{19}{13}$
(B) $\frac{40}{27}$
(C) $\frac{1}{3}$
(D) $\frac{1}{4}$

Q14 JEE Main 2020 - 9 January (Morning)

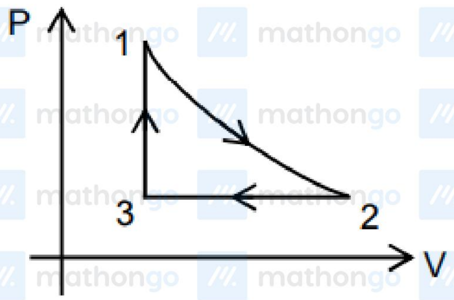
Consider two ideal diatomic gases A and B at some temperature T. Molecules of the gas A are rigid, and have a mass m. Molecules of the gas B have an additional vibrational mode, and have a mass $4m$. The ratio of the specific heats $({C_v}^A \; and \; {C_v}^B)$ of gas A and B, respectively is :

(A) $\frac{7}{9}$

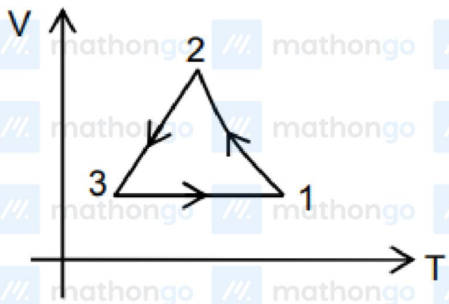
(B) $\frac{5}{9}$
(C) $\frac{5}{11}$
(D) $\frac{5}{7}$

Q15 JEE Main 2020 - 9 January (Morning)

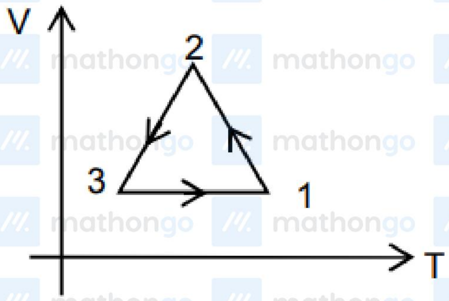
For the given P-V graph for an ideal gas, chose the correct V-T graph. Process BC is adiabatic.



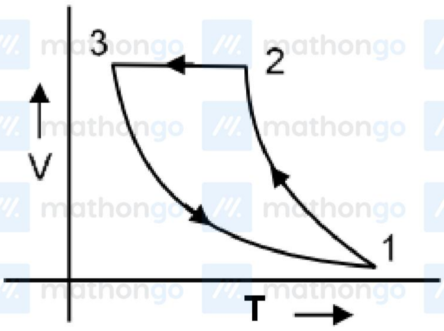
(A)



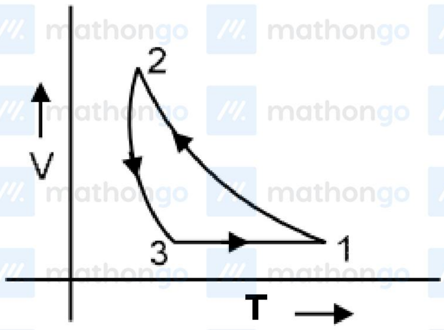
(B)



(C)



(D)



Q16 JEE Main 2020 - 9 January (Evening)

Starting at temperature $300\ K$, one mole of an ideal diatomic gas $(\gamma = 1.4)$ is first compressed adiabatically from volume V_1 to $2V_2 = V_1/16$. it is then a allowed to expand isobaric-ally to volume $2V_2$. If all the processes are the quasi-static then the final temperature of the gas (in $^\circ K$) is (to the nearest integer)

Answer Key

Q1 (C)	**Q2** (266.67)	**Q3** (150)	**Q4** (D)
Q5 (40.93)	**Q6** (5)	**Q7** (C)	**Q8** (A)
Q9 (B)	**Q10** (B)	**Q11** (A)	**Q12** (A)
Q13 (A)	**Q14** (D)	**Q15** (A)	**Q16** (1819)

Q1

Mean free path is independent of temperature and relaxation time decreases as temperature increases.

Q2

$$\left(0.1 \times \frac{3}{2} R \times 200\right)+\left(0.05 \times \frac{3}{2} R \times 400\right)=0.15 \times \frac{3}{2} R \cdot T_f$$
$$\Rightarrow(20+20)=0.15 T_f$$
$$\Rightarrow T_f=\frac{40}{15} \times 100 \simeq 266.67$$

Q3

For temperature to be constant

$$\mathrm{PV}=\mathrm{NRT}$$
$$\Rightarrow \mathrm{PdV}+\mathrm{VdP}=0$$
$$\Rightarrow|\Delta \mathbf{V}|=\left|\frac{\mathrm{VdP}}{\mathbf{P}}\right|$$

And at constant pressure

$$\mathrm{PdV}=\mathrm{NRdT} \quad \Rightarrow|\mathrm{dV}|=\left|\frac{\mathrm{NRdT}}{\mathrm{P}}\right|$$
$$\frac{V d P}{P}=\frac{N R d T}{P} \quad \Rightarrow V d P=N R C d P$$
$$\therefore \quad \mathrm{V}=\mathrm{NRC} \quad \Rightarrow \mathrm{C}=\frac{\mathrm{V}}{\mathrm{NR}}=\frac{\mathrm{T}}{\mathrm{P}}$$
$$\Rightarrow \mathrm{C}=\frac{300 \mathrm{K}}{2 \mathrm{atm}}=150\left(\frac{\mathrm{K}}{\mathrm{atm}}\right)$$

Q4

PV = NkT

$$\mathrm{E}=\frac{3}{2} \mathrm{kT}$$
$$\mathrm{N}=\frac{3 \mathrm{PV}}{2 \mathrm{E}} \quad(\mathrm{P}=\mathrm{h} \rho \mathrm{g})$$
$$=4 \times 10^{18}$$

Q5

V_{rms} for $N_2 = \sqrt{\frac{3KT}{28}} = \sqrt{\frac{3\times K\times 573}{28}}$

v_{rms} for $H_2 = \sqrt{\frac{3KT_1}{2}}$

$\because v_{rms}$ for $N_2 = V_{rms}$ for H_2

$\therefore \quad \frac{3KT_1}{2} = \frac{3K\times 573}{28} \Rightarrow T_1 = \frac{573}{14}K$

$T_1 = 40.93\text{kelvin}$

Q6

$n_0 = \frac{P_1V_1}{R\times 250}$

$n' = 0.75n_0 + 0.5n_0$

$= 1.25n_0$ moles

$P_2 \times 2V_1 = (1.25)\frac{P_1V_1}{R\times 250} \times R \times 2000$

$\Rightarrow \frac{P_2}{P_1} = 5$

Q7

$\tau \propto \frac{1}{\sqrt{T}}$

$\because \ell_{mean} = \frac{1}{\sqrt{2}\pi n D^2}$ and $v \propto \sqrt{T}$

$\therefore \quad \tau_{mean} = \frac{\ell_{mean}}{v}$

$\Rightarrow \quad \top \propto \frac{1}{\sqrt{T}}$

Q8

$$V_{\mathrm{rms}}=\sqrt{\frac{3\mathrm{KT}}{\mathrm{m}}}$$
$$\lambda=\frac{\mathbf{h}}{\sqrt{3\mathrm{KTm}}}$$

Q9

$$\gamma_{mixture}=\frac{n_1C_{P_1}+n_2C_{P_2}}{n_1C_{V_1}+n_2C_{V_2}}=\frac{n_1\frac{\gamma_1R}{\gamma_1-1}+n_2\frac{\gamma_2R}{\gamma_2-1}}{\frac{n_1R}{\gamma_1-1}+\frac{n_2R}{\gamma_2-1}}$$

on rearranging we get

$$\frac{n_1+n_2}{\gamma_{mix}-1}=\frac{n_1}{\gamma_1-1}+\frac{n_2}{\gamma_2-1};\ \frac{5}{\gamma_{mix}-1}=\frac{3}{1/3}+\frac{2}{2/3}$$
$$\frac{5}{\gamma_{mix}-1}=9+3=12\ \Rightarrow\ \gamma_{mixture}=\frac{17}{12}=1+\frac{5}{12};$$
$$\gamma_{mix}=1.42$$

Q10

$$P_1=1\text{ atm}, T_1=273\ K$$
$$P_1V_1^{\gamma}=P_2V_2^{\gamma}$$
$$P_2=P_1\left[\frac{V_1}{V_2}\right]^{\gamma}$$
$$=1\text{ atm}\left(\frac{1}{3}\right)^{1.4}$$

$$\text{now work done}=\frac{P_1V_1-P_2V_2}{\gamma-1}=88.7\ J$$

Q11

$$\tau\propto\frac{1}{\sqrt{T}}$$

Q12

Self-Explanatory

Q13

$$\gamma_{mix} = \frac{n_1 c_{p_1} + n_2\, c_{p_2}}{n_1 c_{v_1} + n_2\, c_{v_2}}$$

$$= \frac{n\left(\frac{5}{2}R\right) + 2n\left(\frac{7}{2}R\right)}{n\left(\frac{3}{2}R\right) + 2n\left(\frac{5}{2}R\right)}$$

$$= \frac{5+14}{3+10} = \frac{19}{13}$$

Q14

Molar heat capacity of A at constant volume $= \dfrac{5R}{2}$

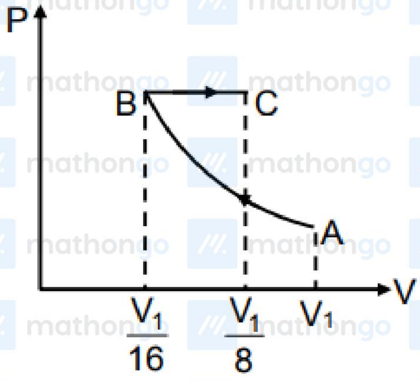
Molar heat capacity of B at constant volume $= \dfrac{7R}{2}$

Dividing both $\dfrac{(C_v)_A}{(C_v)_B} = \dfrac{5}{7}$

Q15

For process $A - B$
Volume is constant
$PV = nRT$; as P increases T increases
For process $B - C$
$PV^{\gamma} = \text{Constant}$
$\Rightarrow \quad TV^{\gamma-1} = \text{Constant}$
For process $C - A$; pressure is constant
$V = kT$

Q16



$PV^y = \text{constant}$
$TV^{y-1} = \text{constant}$

$$300(V_1)^{1.4-1} = T_B\left(\frac{V_1}{16}\right)^{2/5}$$

$T_B = 300 \times 2^{8/5}$
Now for BC process

$$\frac{V_B}{T_B} = \frac{V_C}{T_C}$$

$$T_C = \frac{V_C T_B}{V_B} = 2 \times 300 \times 2^{8/5}$$

$T_C = 1818.859$
$T_C = 1819\ K$